

LIST OF MATHEMATICAL FORMULAE

Trigonometry

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Differentiation

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

Product Rule

$$\frac{d}{dx}(u(x)v(x)) = v \frac{du}{dx} + u \frac{dv}{dx}$$

Quotient Rule

$$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain Rule

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

Shape	Volume	Surface Area
Sphere	$V = \frac{4}{3} \pi r^3$	$V = 4\pi r^2$
Right circular cone	$V = \frac{1}{3} \pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

SECTION A [25 MARKS]

This section consists of **3** questions. Answer **all** questions.

1. Evaluate.

(a) $\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 6}$ [4 marks]

(b) $\lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$ [5 marks]

2. (a) If $y = \sin(x^2 + 1)$, show that $x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = 0$. [5 marks]

(b) Find the first derivative of $e^{3xy} = 11x - 5y$. [4 marks]

3. By considering the first and second derivatives of $f(x)$, show that there is only one maximum

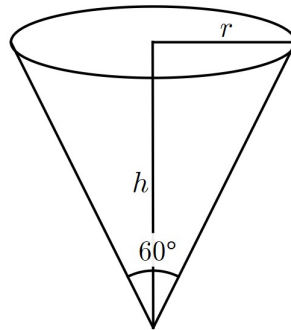
point on the graph $f(x) = \frac{\ln(x-1)}{x-1}$. [7 marks]

SECTION B [75 MARKS]

This section consists of 7 questions. Answer **all** questions.

1. Solve for p and q , where $p \neq q$, such that $\frac{p+qi}{3i} = 3 + \sqrt{-16} - i^3$. [6 marks]
2. (a) If $\sqrt{7-3\sqrt{5}} = \sqrt{x} - \sqrt{y}$, determine the values of x and y . [6 marks]
(b) Find the solution set for the inequality $\frac{-4-x}{x-3} \geq x+4$. [6 marks]
3. Given $g(x) = \sqrt[3]{x}$ and $h(x) = \frac{1}{x^3}$. Find $f(x)$ such that $(f \circ g \circ h)(x) = \frac{x}{x+1}$. [4 marks]
4. Given that $f(x) = e^{-2x} + 3$ and $(g \circ f)(x) = \frac{1}{2} \ln(e^{-2x} + 2)$.
(a) Find domain and range of $f(x)$. [3 marks]
(b) Find $g(x)$. [5 marks]
(c) Hence, find $g^{-1}(x)$. Sketch the graph of $g(x)$ and $g^{-1}(x)$ on the same axes. [6 marks]
5. Given that $f(x) = \frac{1-x^2}{2x+1}$.
(a) Show that the $f(x)$ has one vertical asymptote and state the equation. [3 marks]
(b) Find the horizontal and vertical asymptotes of the function $g(x) = \frac{1}{f(x)}$. [8 marks]
(c) Sketch the graph of $g(x)$. [3 marks]
6. (a) If $y = e^x \ln(1+x)$, show that $(x+1)^2 \left(\frac{d^2y}{dx^2} - \frac{dy}{dx} \right) = xe^x$. [6 marks]
(b) Determine whether the function $f(x) = \begin{cases} x^2, & x \leq 1, \\ x+1, & x > 1, \end{cases}$ is differentiable at $x=1$. [5 marks]
(c) Find $\frac{dy}{dx}$ for $y = x^{\sin x}$. [4 marks]

7. (a) Use the derivative to find the maximum area of a rectangle that can be inscribed in a semicircle of radius 10 cm. [5 marks]
- (b)



Water drains from the conical tank shown at a rate of 6 cm^3 per second. Find the rate of change in height of the water when the volume of water in the cone is $24\pi \text{ cm}^3$. [5 marks]

END OF QUESTION PAPER

Answers

SECTION A

1. (a) $\sqrt{3}$ (b) $\frac{3}{2}$

2. (a) *DIY* (b) $\frac{dy}{dx} = \frac{11 - 3ye^{3xy}}{3xe^{3xy} + 5}$

3. *DIY*

SECTION B

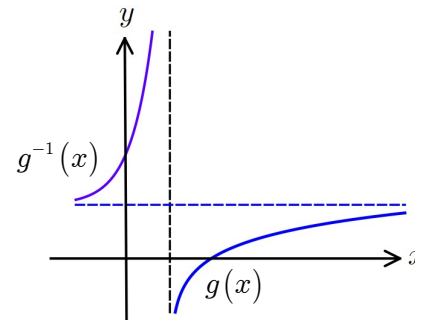
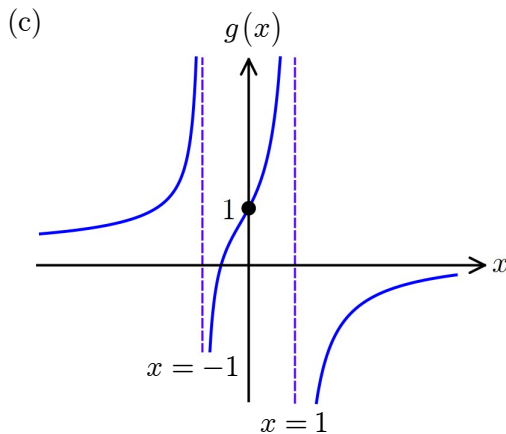
1. $p = -15, q = 9$

2. (a) $x = \frac{9}{2}, y = \frac{5}{2}$ (b) $\{x : x \leq -4 \text{ or } 2 \leq x < 3\}$

3. $f(x) = \frac{1}{1+x}$

4. (a) $D_f = \mathbb{R}, R_f = (3, \infty)$ (b) $g(x) = \frac{1}{2} \ln(x-1)$ (c) $g^{-1}(x) = e^{2x} + 1$

5. (a) VA: $x = -\frac{1}{2}$ (b) HA: $y = 0$, VA: $x = -1, x = 1$



6. (a) *DIY* (b) f is not differentiable at $x = 1$ (c) $\frac{dy}{dx} = x^{\sin x} \left[\sin x \left(\frac{1}{x} \right) + \ln x (\cos x) \right]$

7. (a) 100 cm^2 (b) $\frac{dh}{dt} = -\frac{1}{2\pi} \text{ cms}^{-1}$